

Engineering and Infrastructure Report - Q3

Infrastructure Investments

The platform team completed a major migration of the primary database cluster from a single-region deployment to a multi-region active-passive configuration during Q3. This migration was originally planned for Q2 but was delayed six weeks after an initial dry run in a staging environment surfaced replication lag issues under peak load. The production cutover occurred over a single weekend in late August with roughly four minutes of read-only downtime and no data loss. Average p99 read latency improved by 34 percent following the migration, primarily due to the new regional read replicas serving traffic closer to end users.

Cloud infrastructure spend for the quarter was 340,000 dollars, up from 298,000 dollars in Q2. The increase is attributable to the new database replicas and to increased compute usage from the analytics dashboard feature, which processes significantly more data per request than the core product surfaces. The infrastructure team estimates that a planned Q4 initiative to move batch analytics processing to spot instances could reduce this category of spend by approximately 15 percent.

Major Feature Launches

From an engineering perspective, the analytics dashboard was the largest technical undertaking of the quarter, requiring a new streaming data pipeline built on top of the existing event bus. The pipeline processes approximately 40 million events per day at current volume. Load testing prior to launch simulated 5x current volume without degradation, which the team considers sufficient headroom through at least the next three quarters of projected growth.

The mobile notification reliability fix shipped in September involved replacing the third-party push notification vendor's SDK with a first-party implementation after root-causing intermittent delivery failures to a race condition in the vendor's background retry logic. The fix required updates to both the iOS and Android clients and was rolled out gradually over two weeks, starting with 5 percent of devices before reaching full rollout.

Technical Debt and Reliability

The team maintained a 99.94 percent uptime for the core API during Q3, against an internal target of 99.9 percent. There was one incident that

breached the internal severity-1 threshold, described in the security and incidents section below. Three severity-2 incidents occurred during the quarter, all related to elevated error rates during the database migration cutover window and resolved within the planned maintenance window.

A technical debt audit conducted in July identified the authentication service as the highest-priority area for refactoring, citing its monolithic structure and lack of test coverage as a growing risk to release velocity. A dedicated two-engineer team was allocated to begin this refactor in Q4, with a target of extracting authentication into an independently deployable service by the end of Q1 next year.

Security Incidents

One severity-1 incident occurred in early September when a misconfigured access control rule, introduced during a routine permissions cleanup, briefly allowed a small subset of internal analytics dashboards to be reachable without authentication for approximately 40 minutes before being detected by automated monitoring and rolled back. An investigation found no evidence that the exposed dashboards were accessed by anyone outside the company during the window. The incident prompted the team to add an automated pre-deploy check that flags any change reducing authentication scope on internet-facing routes, which has been in place since mid-September.

No other security incidents meeting the internal disclosure threshold occurred during the quarter. The annual third-party penetration test was conducted in August and identified two medium-severity findings, both related to verbose error messages in a deprecated internal API; both were remediated within the test provider's recommended 30-day window.

Team Structure and Hiring

Engineering headcount stood at 58 at quarter end, unchanged from Q2, as noted in the business review. Two senior backend engineering hires originally expected to start in Q3 had their start dates pushed into early Q4 due to visa processing delays, which contributed to the database migration timeline slipping from the original Q2 target. The platform, product, and mobile sub-teams remained at their Q2 staffing levels throughout the quarter with no attrition.

Third-Party Vendor Changes

Beyond the push notification vendor replacement described above, the team completed an evaluation of the primary error-monitoring vendor during Q3 and elected to remain on the incumbent provider after a competing vendor's pricing proposal came in only marginally lower once the cost of a migration was factored in. The team also onboarded a new log-aggregation vendor in September to replace a self-hosted solution that had become increasingly costly to operate at current data volumes; this migration is expected to reduce related infrastructure spend by roughly 20,000 dollars per quarter going forward.

Upcoming Roadmap Q4

Q4 engineering priorities are, in order: completing the authentication service refactor kickoff, shipping the spot-instance migration for batch analytics processing, and beginning discovery work on a public API for enterprise customers, which has been the most frequently requested feature in the enterprise sales pipeline over the past two quarters. The team also plans to complete a second load-testing pass on the analytics pipeline ahead of an anticipated Q1 marketing push expected to meaningfully increase signup volume.

Cost Optimization Initiatives

In addition to the planned spot-instance migration, the infrastructure team identified approximately 45,000 dollars per quarter in savings from rightsizing over-provisioned staging environments, a change that was implemented in late September and is expected to be fully reflected in Q4 spend. A broader review of reserved-instance commitments is planned for Q4 as several existing commitments come up for renewal.